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Table 4.2: Differences between male and female cockroach

S. No	Character	Male cockroach	Female cockroach
1.	Abdomen	Long and narrow	Short and broad
2.	Segments	In the abdomen, nine segments are visible	In the abdomen, seven segments are visible
3.	Anal styles	Present	Absent
4.	Terga	7th tergum covers 8th tergum	7th tergum covers 8th and 9th terga
5.	Brood pouch	Absent	Present
6.	Antenna	Longer in length	Shorter in length
7.	Wings	Extends beyond the tip of abdomen	Extends up to the end of abdomen

Anatomy

Digestive system

The digestive system of cockroach consists of the alimentary canal and digestive glands. The alimentary canal is present in the body cavity and is divided into three regions: foregut, midgut and hindgut (Figure 4.10). The foregut includes pre-oral cavity, mouth, pharynx and oesophagus. This in turn opens into a sac like structure called the **crop** which is used for storing food. The crop is followed by the **gizzard** or **proventriculus** which has an outer layer of thick circular muscles and thick inner cuticle forming six highly chitinous plates called **"teeth"**. Gizzard helps in the grinding of the food particles. The midgut is a short and narrow tube behind the gizzard and is glandular in nature. At the junctional region of the gizzard are eight fingers like tubular blind processes called the **hapatic caecae** or **enteric caecae.** The hindgut is marked by the presence of 100 – 150 yellow coloured thin filamentous **malphigian** tubules which are helpful in removal of the excretory products from the haemolymph. The hindgut is broader than the midgut and is differentiated into ileum, colon, and rectum. The rectum opens out through the anus.

Digestive glands of cockroach consist of the salivary glands, the glandular cells and hepatic caecae. A pair of salivary glands is found on either side of the crop

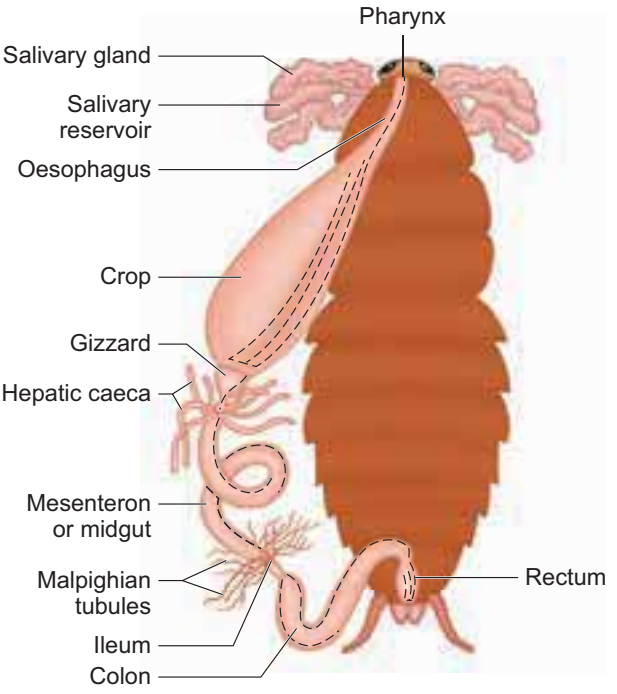


Figure 4.10 *Periplaneta americana*: Digestive system